

Chapter 9b Practice Test - #1 – 9 NC (25 min), #10 – 16 CA (30 min)

1. What are the first four nonzero terms in the power series expansion of e^{-4x} about $x = 0$?

a) $1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ b) $1 - 4x + 8x^2 - 32x^3$ c) $1 - 4x - 2x^2 - \frac{2}{3}x^3$
 d) $1 - 4x + 8x^2 - \frac{32}{3}x^3$ e) $1 - 4x + 8x^2 - \frac{64}{3}x^3$

2. What is the third degree term in the Taylor series for $y = \sin(\pi x) - x^2$ at $c = \frac{1}{3}$?

a) $\frac{-\pi^3}{2} \left(x - \frac{1}{3}\right)^3$ b) $\frac{-\pi^3}{2} x^3$ c) $\frac{\pi^3}{6} \left(x - \frac{1}{3}\right)^3$ d) $\frac{-\pi^3}{12} \left(x - \frac{1}{3}\right)^3$ e) $\frac{-\pi^3}{3} \left(x - \frac{1}{3}\right)^3$

3. At what value of x does the function $y = f(x)$ represented by the series $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ reach a y -value of $\frac{1}{2}$?

a) 0 b) $\frac{\pi}{6}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{3}$ e) $\frac{\pi}{2}$

4. What is the radius of convergence of the series $\sum_{n=0}^{\infty} \frac{(x-3)^{2n}}{(n+1)2^n}$?

a) 4 b) 3 c) 2 d) $\sqrt{2}$ e) 1

5. Find the series, centered at $c = -1$, for the function $\frac{6}{4+x}$.

a) $\sum_{n=0}^{\infty} \frac{3}{2} \left(\frac{x}{4}\right)^n$ b) $\sum_{n=0}^{\infty} \frac{3}{2} \left(-\frac{x}{4}\right)^n$ c) $\sum_{n=0}^{\infty} 2 \left(\frac{x+1}{3}\right)^n$ d) $\sum_{n=0}^{\infty} 2(-1)^n \left(\frac{x+1}{3}\right)^n$ e) $\sum_{n=0}^{\infty} 6(-1)^n (x+1)^n$

6. Let $P(x) = 3x^2 - 5x^3 + 7x^4 + 3x^5$ be the fifth-degree Taylor polynomial for the function f about $x = 0$. What is the value of $f'''(0)$?
- a) -30 b) -15 c) -5 d) $-\frac{5}{6}$ e) $-\frac{1}{6}$
7. Given $f'(x) = x^2 \cos x$, which of the following represents the Maclaurin power series for $f(x)$?
- a) $2x - \frac{4x^3}{2!} + \frac{6x^5}{4!} - \frac{8x^7}{6!} + \dots$ b) $-\frac{x^3}{3} + \frac{x^5}{5(2!)} - \frac{x^7}{7(4!)} + \frac{x^9}{9(6!)} - \dots$ c) $\frac{x^3}{3} - \frac{x^5}{5(2!)} + \frac{x^7}{7(4!)} - \frac{x^9}{9(6!)} + \dots$
d) $\frac{x^4}{4} - \frac{x^6}{6(3!)} + \frac{x^8}{8(5!)} - \frac{x^{10}}{10(7!)} + \dots$ e) $x^2 - \frac{x^4}{2!} + \frac{x^6}{4!} - \frac{x^8}{6!} + \dots$
8. The interval of convergence for the power series $\sum_{n=0}^{\infty} (\cos \pi n)(x+2)^n$ is...
- a) $-3 < x < -1$ b) $-3 < x \leq -1$ c) $-1 < x \leq 1$ d) $-1 < x < 1$ e) $-1 < x < 3$
9. Simplify the following: $1 - \frac{3^2}{2} + \frac{3^4}{24} - \frac{3^6}{720} + \dots$
- a) 0 b) $\sin 3$ c) $3e$ d) e^3 e) $\cos 3$
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10. The Maclaurin series for $\frac{1}{1-x}$ is $\sum_{n=0}^{\infty} x^n$. Which of the following is a power series expansion for $\frac{x^2}{1-x^2}$?
- a) $1 + x^2 + x^4 + x^6 + x^8 + \dots$ b) $x^2 + x^3 + x^4 + x^5 + \dots$ c) $x^2 + 2x^3 + 3x^4 + 4x^5 + \dots$
d) $x^2 + x^4 + x^6 + x^8 + \dots$ e) $x^2 - x^4 + x^6 - x^8 + \dots$
11. Which of the following functions is represented by the power series $3 - 6x^3 + 12x^6 - 24x^9 + 48x^{12} - \dots$?
- a) $\frac{-3}{1-2x^3}$ b) $\frac{-3}{1+2x^3}$ c) $\frac{3}{1-2x^3}$ d) $\frac{3}{1+2x^3}$ e) $\frac{3x}{1+2x^3}$

12. Which of the following is the series expansion for $f'(x)$ given $f(x) = \frac{\sin x}{x}$?

- a) $-\frac{2x}{3!} + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \dots$ b) $\frac{2x}{3!} - \frac{4x^3}{5!} + \frac{6x^5}{7!} - \dots$ c) $-\frac{3x^2}{3!} + \frac{5x^4}{5!} - \frac{7x^6}{7!} + \dots$
 d) $\frac{3x^2}{3!} - \frac{5x^4}{5!} + \frac{7x^6}{7!} + \dots$ e) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

13. The set of all x for which the power series $\sum_{n=0}^{\infty} \frac{x^n}{(n+4)3^n}$ converges is

- a) $-3 \leq x \leq 3$ b) $-3 < x < 3$ c) $x < -3$ or $x > 3$ d) $-3 \leq x < 3$ e) $-3 < x \leq 3$

14. The graph of the function represented by the Maclaurin series $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots + \frac{(-x)^n}{n!} + \dots$ intersects the graph of $y = x^3$ at which of the following value of x ?

- a) 0.773 b) 0.865 c) 0.929 d) 1 e) 1.857

15. A function f has Maclaurin series given by $\frac{x^4}{2!} + \frac{x^5}{3!} + \frac{x^6}{4!} + \dots + \frac{x^{n+3}}{(n+1)!} + \dots$. Which of the following is an expression for $f(x)$.

- a) $-3x \sin x + 3x^2$ b) $-\cos(x^2) + 1$ c) $-x^2 \cos x + x^2$ d) $x^2 e^x - x^3 - x^2$ e) $e^{x^2} - x^2 - 1$

16. Determine the maximum error for approximating $\cos 1.5$ using the first 3 nonzero terms of the Maclaurin series for $\cos x$.

- a) 0.00254 b) 0.00463 c) 0.0063 d) 0.0088 e) 0.01582

17. Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (x-1)^{n+2}}{n+1}$.

a) Find the series for $\int f(x)dx$.

b) Find the interval of convergence for $\int f(x)dx$. Justify your answer.

18. Let $f(x)$ be the function defined by $f(x) = \frac{1}{1-2x}$.

a) Write the first four terms and the general term of the Taylor series expansion of $f(x)$ about $x=0$.

b) Use the first four terms of the polynomial from part a) to approximate $f(0.2)$. Find the maximum error of this approximation given $f^{(4)}(x) = \frac{-384}{(1-2x)^5}$.

c) What is the interval of convergence for the series found in part a)? Justify your answer.

19. The Maclaurin Series for $f(x)$ is given by: $\frac{1}{2!} - \frac{x^2}{4!} + \frac{x^4}{6!} + \dots + \frac{(-1)^n x^{2n}}{(2n+2)!} + \dots$

a) For what values of x does the given series converge?

b) Let $g(x) = 1 - x^2 f(x)$. Write the Maclaurin series for $g(x)$, showing the first three nonzero terms and the general term.

c) Use the first three nonzero terms from part b) to approximate $g(0.2)$. Show that the difference between this approximation and the actual value is less than 1×10^{-7} . Justify your answer.

Answers

1. We can use the Maclaurin series for e^x to find the series for e^{-4x} . So, $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$ and

replacing all x with $-4x$ gives us $e^{-4x} = 1 + (-4x) + \frac{(-4x)^2}{2} + \frac{(-4x)^3}{6} + \dots$ or

$$e^{-4x} = 1 - 4x + \frac{16x^2}{2} - \frac{64x^3}{6} + \dots \text{ or } e^{-4x} = 1 - 4x + 8x^2 - \frac{32x^3}{3} + \dots \text{ so D.}$$

2. We have to use the general form for this Taylor series because this is not Maclaurin. Since we only care

about the third term, we will focus on the third degree general term $\frac{f'''(\frac{1}{3})}{3!} \left(x - \frac{1}{3}\right)^3$. So to find $f'''(\frac{1}{3})$,

we will need 3 derivatives, so $f = \sin \pi x - x^2$, $f' = \pi \cos \pi x - 2x$, $f'' = -\pi^2 \sin \pi x - 2$, $f''' = -\pi^3 \cos \pi x$

and $f'''(\frac{1}{3}) = -\pi^3 \cos \frac{\pi}{3} = \frac{-\pi^3}{2}$. So the third degree term is $\frac{-\pi^3}{3!} \left(x - \frac{1}{3}\right)^3$ or $\frac{-\pi^3}{12} \left(x - \frac{1}{3}\right)^3$ so D.

3. We can see that $y = f(x) = \sin x$ from the expansion. So, this is wondering when $\frac{1}{2} = \sin x$, so $\frac{\pi}{6}$ or B.

4. The radius of convergence can be determined by finding the interval of convergence. Using the Ratio

Test, $\lim_{n \rightarrow \infty} \left| \frac{(x-3)^{2(n+1)}}{((n+1)+1)2^{n+1}} \cdot \frac{(n+1)2^n}{(x-3)^{2n}} \right| < 1$, $\lim_{n \rightarrow \infty} \left| \frac{(x-3)^{2n+2}}{(n+2)2^{n+1}} \cdot \frac{(n+1)2^n}{(x-3)^{2n}} \right| < 1$, $\lim_{n \rightarrow \infty} \left| \frac{(x-3)^2}{2} \cdot \frac{(n+1)}{(n+2)} \right| < 1$. This limit

as $n \rightarrow \infty$ causes the right quotient to go to 1. This gives us $\left| \frac{(x-3)^2}{2} \right| < 1$, $-1 < \frac{(x-3)^2}{2} < 1$ or

$-2 < (x-3)^2 < 2$, $-\sqrt{2} < x-3 < \sqrt{2}$. This gives us $3-\sqrt{2} < x < 3+\sqrt{2}$, so the center is 3 and the radius of convergence is $\sqrt{2}$ or D.

5. This function is set up to be geometric, which means we want to get it into $\sum_{n=0}^{\infty} a(r)^n = \frac{a}{1-r}$ form. Since

this is centered $c = -1$, the function can be rewritten as $\frac{6}{4+x} = \frac{6}{4-1+(x+1)} = \frac{6}{3+(x+1)}$. Since we need

the $1-r$, we can write this as $\frac{2}{1+\left(\frac{x+1}{3}\right)} = \frac{2}{1-\left(-\frac{x+1}{3}\right)}$. This is now geometric, so

$$\frac{6}{4+x} = \frac{2}{1-\left(-\frac{x+1}{3}\right)} = \sum_{n=0}^{\infty} 2\left(-\frac{x+1}{3}\right)^n, \text{ which by pulling out the alternating term, gives us D.}$$

6. There are 2 ways to approach this problem. One is to take 3 derivatives and then plug in 0. The other is to look at the third term of the Taylor polynomial and compare it to the general term. Knowing that the third term of the polynomial is $-5x^3 = \frac{f'''(0)}{3!}x^3$ or $-5x^3 = \frac{f'''(0)}{6}x^3$. Solving for $f'''(0)$ gives us $-30x^3 = f'''(0)x^3$, $f'''(0) = -30$ or **A**.
7. Instead of integrating the function $f'(x)$ to get $f(x)$, which would require integration by parts, we can write the power series for $f'(x)$ and then integrate each of the terms to get $f(x)$. So, knowing that $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$, we can say $x^2 \cos x = x^2 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right)$ or
- $$f'(x) = x^2 \cos x = x^2 - \frac{x^4}{2!} + \frac{x^6}{4!} - \frac{x^8}{6!} + \dots$$
- So, we must integrate these terms to find the terms of $f(x)$.
- This gives us $\int f'(x) dx = \int \left[x^2 - \frac{x^4}{2!} + \frac{x^6}{4!} - \frac{x^8}{6!} + \dots \right] dx = \frac{x^3}{3} - \frac{x^5}{5(2!)} + \frac{x^7}{7(4!)} - \frac{x^9}{9(6!)} + \dots$ or **C**.
8. The first part is an alternating term $\cos(\pi) = (-1)^n$. So the interval of convergence can be found using the Ratio test with $\lim_{n \rightarrow \infty} \left| \frac{(x+2)^{n+1}}{(x+2)^n} \right| < 1$, $\lim_{n \rightarrow \infty} |x+2| < 1$ so this limit leaves us with $|x+2| < 1$. Solving this gives us $-1 < x+2 < 1$, $-3 < x < -1$. We now must check these endpoints. With $x = -3$, $\sum_{n=0}^{\infty} (-1)^n (-1)^n$, which diverges by the nth term test and with $x = -1$, $\sum_{n=0}^{\infty} (-1)^n (1)^n$, which also diverges by the nth term test. Therefore, the interval of convergence is $-3 < x < -1$ or **A**.
9. The denominators can be rewritten as $1 - \frac{3^2}{2!} + \frac{3^4}{4!} - \frac{3^6}{6!} + \dots$ indicating that this is a form of $\cos x$. Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$, we see the x has been replaced with a 3, so this is $\cos 3$ or **E**.
10. The Maclaurin series for $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$. The Maclaurin series $\frac{x^2}{1-x^2}$ is geometric with $a = x^2$, $r = x^2$, which can be written as $\frac{x^2}{1-x^2} = \sum_{n=0}^{\infty} x^2 (x^2)^n = x^2 + x^4 + x^6 + x^8 + \dots$ or **D**.
11. Each term appears to be created by multiplying by the same value, making this geometric. This gives us $a = 3$, $r = -2x^3$ and the function would be $\frac{3}{1 - (-2x^3)} = \frac{3}{1 + 2x^3}$ or **D**.

12. We will use the series for $\sin x$ to create $f(x)$ by dividing by x . So, $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ and

$$f(x) = \frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots. \text{ Taking the derivative gives us } f'(x) = -\frac{2x}{3!} + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \dots \text{ or A.}$$

13. Using the Ratio test to find the interval of convergence gives us $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+5)3^{n+1}} \cdot \frac{(n+4)3^n}{x^n} \right| < 1$, so

$$\lim_{n \rightarrow \infty} \left| \frac{x \cdot x^n}{(n+5)3 \cdot 3^n} \cdot \frac{(n+4)3^n}{x^n} \right| < 1, \lim_{n \rightarrow \infty} \left| \frac{x}{3} \cdot \frac{(n+4)}{(n+5)} \right| < 1. \text{ Evaluating this limit gives us } \left| \frac{x}{3} \right| < 1, -1 < \frac{x}{3} < 1 \text{ or}$$

$$-3 < x < 3. \text{ We then must test the endpoints, so when } x = -3, \text{ the series is } \sum_{n=0}^{\infty} \frac{(-3)^n}{(n+4)3^n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+4)}.$$

This series can be compared to the alternating harmonic series, which converges. Checking when $x = 3$,

$$\text{the series is } \sum_{n=0}^{\infty} \frac{(3)^n}{(n+4)3^n} = \sum_{n=0}^{\infty} \frac{1}{(n+4)}. \text{ This series can be compared to the harmonic series, which}$$

diverges. Therefore, the interval of convergence is $[-3, 3)$ or D.

14. To find the intersection, we should write the Maclaurin series as its equivalent function. The

denominators show that this is a form of e^x . Since $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, we can see that the x has

been replaced with $-x$, so the function is e^{-x} . So, we can allow the calculator to find when $e^{-x} = x^3$, which occurs at $x = 0.773$ or A.

15. Looking at the denominators, we can see that this is a form of e^x . Since $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$, we

can see that the powers have been increased by 2, meaning that e^x is multiplied by $x^2 e^x$, which gives us

$$x^2 e^x = x^2 + x^3 + \frac{x^4}{2!} + \frac{x^5}{3!} + \frac{x^6}{4!} + \dots. \text{ We can then see that the first two terms of this series are removed to}$$

give us f , so we can subtract both x^2 and x^3 . This gives us $f = x^2 e^x - x^3 - x^2$ or D.

16. Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$ is an **alternating series whose terms decrease in absolute value to 0**,

the maximum error will be found using the first omitted/unused term. Using the first 3 terms means that

$$\text{the } -\frac{x^6}{6!} \text{ term is used. So, the maximum error is } \left| -\frac{(1.5)^6}{6!} \right| = 0.01582 \text{ or E.}$$

17. a) Considering the alternating term and the denominator are constants, the function $f(x)$ can be

integrated using u-substitution and a power rule. So, $\int f(x)dx = \frac{(-1)^{n+1}}{n+1} \int \sum_{n=0}^{\infty} (x-1)^{n+2} dx$ with

$$u = x-1, du = dx. \text{ So, } \int f(x)dx = \frac{(-1)^{n+1}}{n+1} \int \sum_{n=0}^{\infty} u^{n+2} du = \frac{(-1)^{n+1}}{n+1} \sum_{n=0}^{\infty} \frac{u^{n+3}}{n+3} \text{ or } \sum_{n=0}^{\infty} \frac{(-1)^{n+1}(x-1)^{n+3}}{(n+1)(n+3)}.$$

b) Using the Ratio test, we get $\lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+4}}{(n+2)(n+4)} \cdot \frac{(n+1)(n+3)}{(x-1)^{n+3}} \right| < 1$, $\lim_{n \rightarrow \infty} \left| \frac{(x-1)(x-1)^{n+3}}{(n+2)(n+4)} \cdot \frac{(n+1)(n+3)}{(x-1)^{n+3}} \right| < 1$

or $\lim_{n \rightarrow \infty} \left| (x-1) \cdot \frac{(n+1)(n+3)}{(n+2)(n+4)} \right| < 1$. This limit goes to 1, so $|x-1| < 1$, $-1 < x-1 < 1$, $0 < x < 2$. We then

must test the endpoints, so when $x = 0$, the series is $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}(-1)^{n+3}}{(n+1)(n+3)} = \sum_{n=0}^{\infty} \frac{(-1)^{2n+4}}{(n+1)(n+3)}$. Since the power is always even, this is not alternating or $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+3)}$. This series can be compared to $\sum_{n=0}^{\infty} \frac{1}{n^2}$,

which converges by p-series. Checking when $x = 2$, the series is $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}(1)^{n+3}}{(n+1)(n+3)} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(n+1)(n+3)}$. This series either converges by the alternating series test or comparing to the already convergent series $\sum_{n=0}^{\infty} \frac{1}{n^2}$, because if the always positive terms converges, the alternating one must also. So, $[0, 2]$.

18. a) This function is setup to be geometric, with $a = 1$, $r = 2x$. This means $f(x) = \sum_{n=0}^{\infty} (2x)^n$ and the first

four terms with the general term are $1 + 2x + 4x^2 + 8x^3 + \dots (2x)^n$.

b) Using the first four terms, $f(x) \approx 1 + 2x + 4x^2 + 8x^3$ and $f(0.2) \approx 1.624$. Since this is not an alternating series, we must use the Lagrange Error bound, stating that the maximum error is

$\left| \frac{f^{(4)}(z)}{4!} x^4 \right|$ where z is some value $0 \leq z \leq 0.2$ that will maximize $|f^{(4)}(x)| = \left| \frac{-384}{(1-2x)^5} \right|$. Using either

a graphing calculator to find the max, or mathematical understanding, we can tell $z = 0.2$. So the

maximum error is $\left| \frac{f^{(4)}(0.2)}{4!} (0.2)^4 \right| = \left| \frac{f^{(4)}(0.2)}{24} (0.2)^4 \right| = \left| \frac{4938.271605}{24} (0.0016) \right| = 0.329218$.

c) Since this series is geometric, the series will only converge if $|r| < 1$, so $|2x| < 1$, $-1 < 2x < 1$ or

$$\left(-\frac{1}{2}, \frac{1}{2} \right).$$

Note, since this is geometric, we know the endpoints will diverge. We could also do the Ratio test and we will get the same answer.

19. a) Using the Ratio test, we get $\lim_{n \rightarrow \infty} \left| \frac{x^{2n+2}}{(2n+4)!} \cdot \frac{(2n+2)!}{x^{2n}} \right| < 1$, $\lim_{n \rightarrow \infty} \left| \frac{x^2 x^{2n}}{(2n+4)(2n+3)(2n+2)!} \cdot \frac{(2n+2)!}{x^{2n}} \right| < 1$ or $\lim_{n \rightarrow \infty} \left| x^2 \cdot \frac{1}{(2n+4)(2n+3)} \right| < 1$. This limit goes to 0, giving us $\lim_{n \rightarrow \infty} |x^2 \cdot 0| < 1$, $0 < 1$, which is always true. Therefore, the interval of convergence is $(-\infty, \infty)$.

b) To construct $g(x)$, we must multiply each term in $f(x)$ by x^2 and then subtract those terms from 1.

So, $x^2 f(x) = x^2 \left(\frac{1}{2!} - \frac{x^2}{4!} + \frac{x^4}{6!} - \dots + \frac{(-1)^n x^{2n}}{(2n+2)!} \right) = \frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots + \frac{(-1)^n x^{2n+2}}{(2n+2)!}$ and subtracting this

from 1 gives $1 - \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \dots + \frac{(-1)^n x^{2n+2}}{(2n+2)!} \right)$ so $g(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.

c) Using the first three terms gives us the approximation $g(0.2) \approx 1 - \frac{(0.2)^2}{2!} + \frac{(0.2)^4}{4!} = 0.98006667$. To

show that the difference between the actual and approximate value is less than 1×10^{-7} , we should find the maximum error and show that this error is less than 1×10^{-7} . Since $g(x)$ is an **alternating series whose terms decrease in absolute value to 0**, the maximum error will be found using the first

omitted/unused term. Using the first 3 terms means that the $-\frac{x^6}{6!}$ term is used. So, the maximum

error is $\left| -\frac{(0.2)^6}{6!} \right| = 8.88888 \times 10^{-8} < 1 \times 10^{-7}$. Since the maximum error is less than 1×10^{-7} , then the

actual difference between $g(0.2)$ and 0.98006667 is less than 1×10^{-7} .